// ENGLISH ASSIGNMENT · PODCAST EPISODE

WANNACRY

THE RANSOMWARE THAT SHOOK THE WORLD.

MAY 12, 2017

200,000+	150+	1 DAY
COMPUTERS	COUNTRIES	GLOBAL OUTBREAK

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PANEL

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INSIDE: Presentation slides + Podcast script



FRIDAY, MAY 12, 2017: ZERO HOUR

200,000+

COMPUTERS INFECTED

150+

COUNTRIES HIT



1 DAY

TO GO GLOBAL

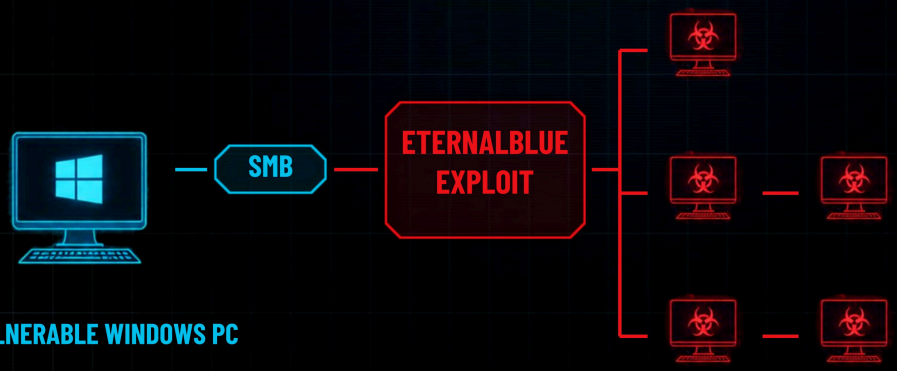
WHAT IS RANSOMWARE?

Malware that locks your files and asks for money to unlock them.
WannaCry encrypted files and showed a red screen demanding Bitcoin.

// 03 · HOW IT SPREAD

ONE VULNERABILITY. A WORM.

SMB is the protocol that lets computers share files on a network.
EternalBlue finds PCs with this weakness.
A worm feature lets it jump to the next computer on its own.
One infected PC can quickly infect every other PC on the network.



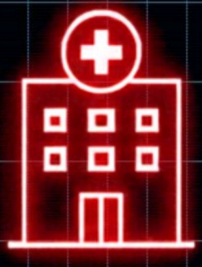
VULNERABLE WINDOWS PC

INFECTED NETWORK NODES
WORM SPREADS AUTOMATICALLY

TOOL LINEAGE — NSA TOOL (ETERNALBLUE) → SHADOW BROKERS LEAK (APRIL 2017) → WANNACRY RANSOMWARE

// 04 · UK NATIONAL HEALTH SERVICE

NHS: CODE RED



UK NATIONAL HEALTH SERVICE

Some of the worst damage was in the UK, where dozens of hospitals were affected.

01 **DOZENS OF HOSPITALS HIT**

Appointments and operations had to be cancelled.

02 **PATIENT RECORDS LOCKED**

Some staff could not access patient records.

03 **AMBULANCES DIVERTED**

Some ambulances had to be sent to other hospitals.

// 05 · BEYOND HEALTHCARE

TELEFONICA

SPAIN · The phone company was affected by the attack.

DEUTSCHE BAHN

GERMANY · Railway information screens were locked.

RENAULT

FEDEX

OLDER WINDOWS = BIGGER DAMAGE

Many hit computers ran older versions of Windows, and many organisations had not updated their systems.

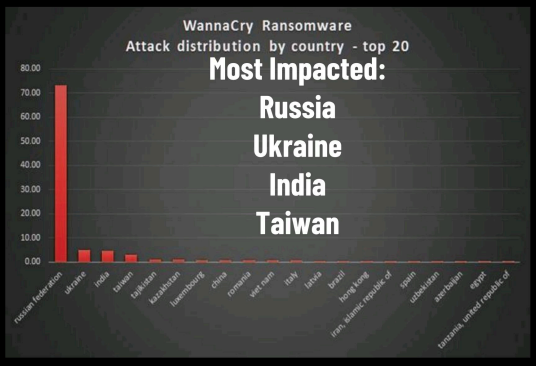
// 06 · WHO WAS HIT

THE OUTBREAK AND ITS VICTIMS

150+

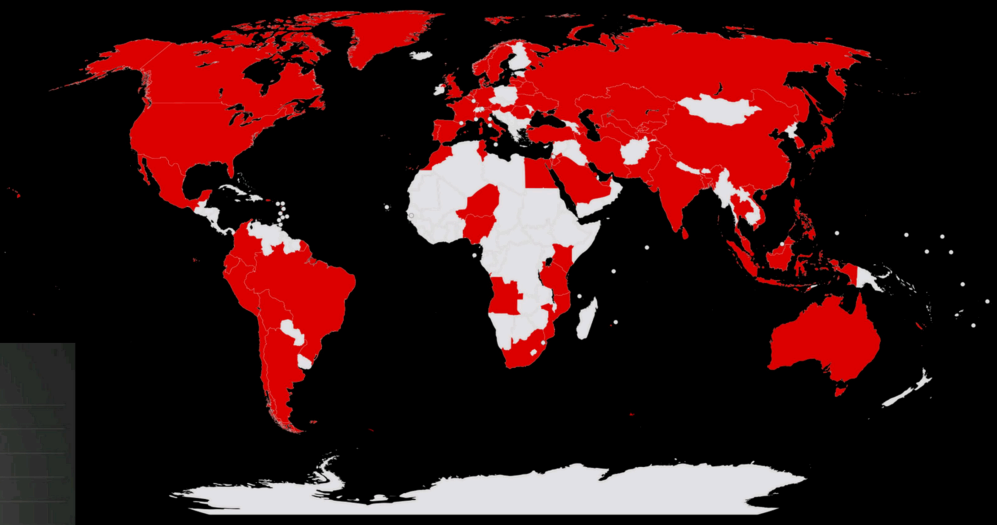
COUNTRIES

AFFECTED
WORLDWIDE



Most Impacted:

Russia
Ukraine
India
Taiwan



AFFECTED ORGANISATIONS

- UK NHS
- TELEFONICA
- DEUTSCHE BAHN
- RENAULT
- FEDEX

// 07 · THE DEMAND

THE RANSOM CLOCK



PAYMENT DEADLINE

$300
BITCOIN RANSOM

Victims were given 3 days to pay.

$600
PRICE DOUBLES

Not paid in time? The price doubled.

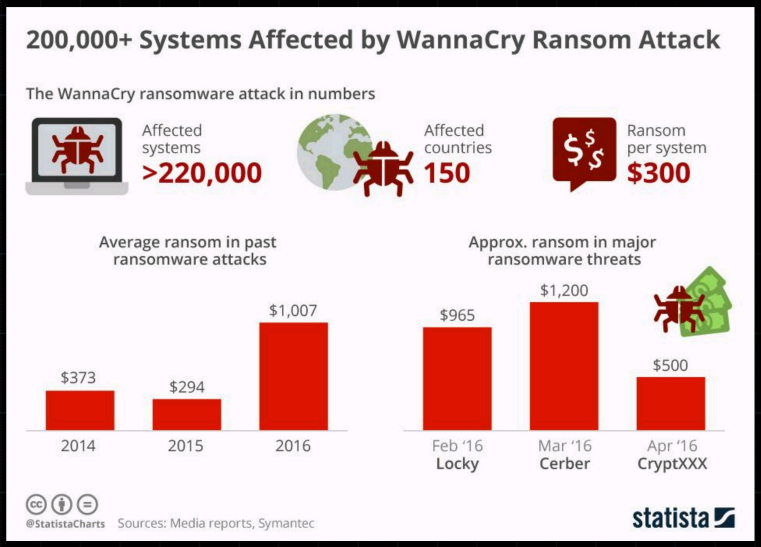
7 DAYS
FILES DELETED?

After a week, files were threatened with deletion.

WHY BITCOIN?

Bitcoin is a digital currency, so it is hard to trace. The ransom was set at $300, with a 3-day deadline.

AROUND $140,000 COLLECTED



LESSER THAN EXPECTED

Compared to other attacks, not much money was asked.

NO GUARANTEE

Paying does not guarantee that you get your files back.

EXPERTS STRONGLY ADVISED: DO NOT PAY

// 09· THE KILL SWITCH

A $10 DOMAIN SLOWED THE SPREAD

Marcus Hutchins, a young British security researcher, saw WannaCry check a web address before encrypting. If the address responds, it **stops**. He registered it for about $10.

MALWARE EXECUTES

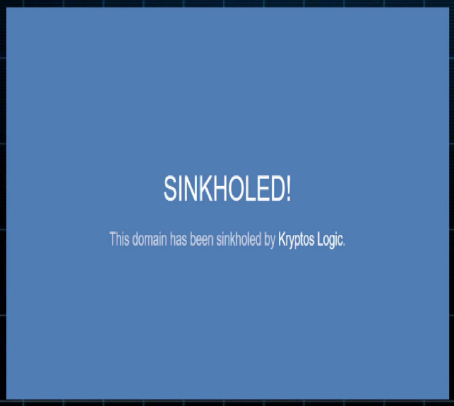
WannaCry starts on a new computer.

→

CHECKS WEB ADDRESS

It looks for one specific domain first.

→



→

STOP SIGNAL

iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com

// 10 · ATTRIBUTION

WHO WAS BEHIND IT?

DEC 2017: OFFICIALLY BLAMED

The United States and United Kingdom named North Korea as responsible.

LAZARUS GROUP LINK

Researchers tied WannaCry to Lazarus, a group connected to the North Korean government. Mistakes in the code also pointed toward a specific group.

NORTH KOREA DENIES

North Korea has denied any involvement.



ATTRIBUTION IS DIFFICULT

Some experts are careful about certainty.

WHAT IS CLEAR

A planned, powerful attack built on tools created by a government agency.

LESSONS LEARNED



01
UPDATE SOFTWARE

Microsoft released a patch in March 2017, almost two months before the attack. Many computers never installed it. A special patch even covered unsupported Windows XP.



02
KEEP BACKUPS

Make regular backups and store a recent copy away from your main computer. Then ransomware is far less dangerous.



03
PLAN AHEAD

Have clear steps for what to do during an attack. One old computer can affect a hospital, a company, even a country.

// 12 · MISSION DEBRIEF

UPDATE. BACK UP. PREPARE.

- Fast-spreading ransomware that hit 150+ countries.
- Made possible by a leaked government tool and an old, unfixed weakness.
- Real harm to hospitals and businesses.
- Slowed by one young researcher who found a kill switch.

NEVER ASSUME IT WON'T HAPPEN TO YOU.



Script

Host(abdullah):

Hello everyone, and welcome to our podcast!

Today, we're going to discuss one of the biggest cyberattacks in history — the WannaCry ransomware attack of 2017.

Imagine switching on your computer and discovering that all your files are locked, and someone is demanding money to unlock them. Sounds scary, right? Well, this is exactly what happened to thousands of people and organizations worldwide.

To explore this incident, we have six panel members with us today: Mr. Krist Allen, Ms. Raksha, Mr. Agnibho, Ms. Lakshitha, Mr. Sanjaysrithan, and Ms. Anika. Our panel contains cyber security analyst and specialist in digital threats . Together, we'll discuss the cyberattack, its global impact, and the lessons we can learn to stay safe in the digital world.

On May 12, 2017, WannaCry began spreading rapidly, infecting more than 200,000 computers across over 150 countries. It locked people's files and demanded payment in Bitcoin.

So Mr. Allen…….

That’s shocking! But here's the big question: how did WannaCry spread so quickly, even without people clicking on suspicious links?

Allen :

The main reason is a weakness in Windows. The weakness was in a protocol called SMB, which helps computers share files on a network. WannaCry used a tool called Eternal Blue to find computers with this weakness. Eternal Blue was originally developed by the United States National Security Agency, the NSA. In April 2017, a hacker group called the Shadow Brokers leaked this tool online. Once it was public, anyone could use it. WannaCry took that tool and added a worm feature. This means it could jump from one computer to another on its own, without anyone clicking anything. One infected computer could quickly infect every other computer on the same network.

Host(abdullah):

We've seen how the attack spread, but who suffered the most? And how did it affect hospitals and major companies? ……..Mr. Agnibho please give your views on this…

Agnibho:

As we can see from the slide, over 150 countries were affected by the WannaCry Ransomware with some of the hardest hit countries being Russia, Ukraine, India and Taiwan. Some of the worst damage was in the United Kingdom. Many hospitals in the National Health Service, the NHS, were affected. Dozens of hospitals had to cancel appointments and operations. Some staff could not access patient records, and some ambulances had to be sent to other hospitals. Outside of healthcare, the attack hit major companies too. In Spain, the phone company Telefonica was affected. In Germany, the railway company Deutsche Bahn showed information screens that were locked. The car company Renault and the delivery company FedEx also reported problems. Many of the computers that were hit were running older versions of Windows. This made the damage worse, because many organizations had not updated their systems.

Host (abdullah):

My next question is from miss raksha……..

The damage was huge, but what did the hackers gain from all this? How much money did they demand, and did people actually pay?

Raksha:
The ransom was set at 300 dollars in Bitcoin. The message gave victims three days to pay. If they did not pay in time, the price would double to 600 dollars. If they waited longer than seven days, the message warned that the files would be deleted. Bitcoin is a digital currency, so it is hard to trace. The hackers only received a small amount of money. Researchers estimate that the total payments were around 140,000 dollars. That is a small amount compared to the damage the attack caused. Security experts strongly advised people not to pay, because paying does not guarantee that you get your files back.

Host (Abdullah):

Now, here's an interesting twist! How was WannaCry finally slowed down, and how did one young researcher make such a big difference?
Miss anika please give your insights on this……

Anika :

This part of the story is very interesting, i think. A 22 year old British security researcher and ex hacker named Marcus Hutchins was looking through the WannaCry code while on vacation, trying to reverse engineer the attack. He found that the program checked a specific web address before it started encrypting files. If the program could reach that address, it would stop. Hutchins registered that domain name for about ten dollars. He turned it into a sinkhole, meaning that it re directed the malicious traffic into a controlled environment and allowed the security professionals at his company to analyse the problem. Because of that, the new copies of WannaCry that tried to reach the address found a working website, and they shut down. While this did not save computers that had already been infected by the ransomware--those would have to update their systems or cut their losses and re set everything--it acted as a kill switch that stopped the spread and saved millions. With the kill switch in place, researchers could set about patching the exploit, and they even discovered a way to unlock and decrypt affected computers without having to pay ransom using a tool called WannaKey, developed in France
While the battle between the hackers and the responders had more complications, their quick thinking and global cooperation brought the whole affair down to something manageable by the 4th day.

Host(Abdullah):

That brings us to another important question:…….Mr. Sanjay! who was behind this massive cyberattack, and how did investigators identify the suspected attackers?

Sanjay :

In December 2017, the United States and the United Kingdom officially blamed North Korea. Security researchers linked WannaCry to a hacking group called the Lazarus Group, which is connected to the North Korean government. North Korea has denied any involvement. Some researchers also found mistakes in the code that pointed toward a specific group. Still, attribution in cyberattacks is very difficult, so some experts are careful about saying anything with full certainty. What is clear is that the attack was not a random accident. It was a planned and powerful attack that used tools created by a government agency.

Host(abdullah):

Finally miss Lakshitha….. what can we learn from WannaCry, and how can we protect ourselves from similar attacks?

Lakshitha:

The first lesson is to keep your software updated. Microsoft had already released a security patch for the weakness in March 2017, almost two months before the attack. Many computers never installed it. Microsoft even released a special patch for Windows XP, an old system that the company had already stopped supporting.The second lesson is to make regular backups. If you have a recent backup stored away from your main computer, a ransomware attack is much less dangerous. Backups allow individuals and organizations to recover important files without depending entirely on the attackers. It is important to keep multiple copies of significant data in separate locations, including offline storage, so that ransomware cannot easily encrypt every copy. The third lesson is to plan ahead. Organizations should have clear steps for what to do during an attack. They should know how to identify infected computers, isolate affected systems, and restore normal operations safely. Employees should also know whom to contact if they notice suspicious activity. WannaCry showed that a problem with one old computer can affect an entire hospital, a company, or even a country. It also demonstrated how quickly ransomware can spread across connected networks.

Host(Abdullah):

And that brings us to the end of today's episode!

WannaCry showed the world that a single weakness in computer security can affect entire businesses, hospitals, and countries.

The lesson is simple: keep your systems updated, back up your important data, and always stay alert online.

Remember, in the digital world, prevention is always better than recovery!

Thank you to all our panel members for sharing their insights, and thank you, everyone, for listening to our podcast.